

CS 237: PROBABILITY IN COMPUTING

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UNIVERSITY



LECTURE 16

Last time

- Law of Total Expectation
- Intro. to Variance

Today

- Variance properties
- Discrete distributions I

<https://www.mathsisfun.com/data/quincunx.html>

RECAP: VARIANCE

- ▶ **Definition:** The variance of a random variable X is

$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2)$$

- ▶ We have also seen that $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$
- ▶ Can compute $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$ from the PDF of X :

discrete: $\mathbb{E}(X) = \sum_{x \in \mathbb{R}} x \cdot f_X(x) \quad \mathbb{E}(X^2) = \sum_{x \in \mathbb{R}} x^2 \cdot f_X(x)$

continuous: $\mathbb{E}(X) = \int_{-\infty}^{\infty} x \cdot f_X(x) dx \quad \mathbb{E}(X^2) = \int_{-\infty}^{\infty} x^2 \cdot f_X(x) dx$

STANDARD DEVIATION

- ▶ **Definition:** The standard deviation of a random variable X is

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{\mathbb{E}((X - \mathbb{E}(X))^2)}.$$

- ▶ Since the units of variance are squared, we consider the square root of the variance.

VARIANCE

- ▶ **Theorem:** $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$

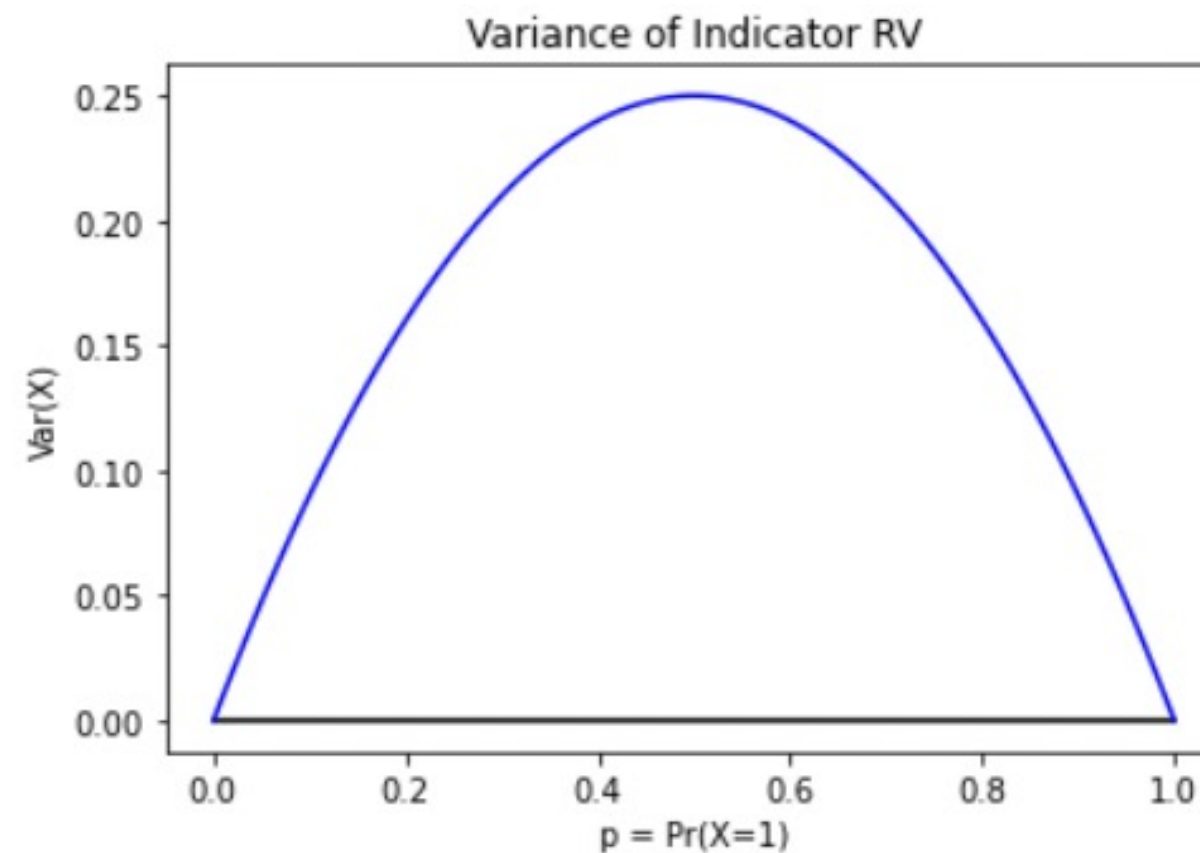
EXAMPLE: COIN TOSS

- ▶ Toss a coin that comes up Heads with probability p
- ▶ Let $X = \begin{cases} 1 & \text{if Heads} \\ 0 & \text{if Tails} \end{cases}$
- ▶ Since X is an indicator random variable, we know $\mathbb{E}(X) = p$
- ▶ $\text{Var}(X) =$

EXAMPLE: COIN TOSS

- ▶ For $p = \frac{1}{2}$

$$\text{Var}(X) = p(1 - p) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$



ASK THE AUDIENCE

Let X be an indicator Random Variable with $\mathbb{P}[X = 1] = 1$

What is $\text{Var}(X)$?

A. 0

B. 1

C. We need more information to determine $\text{Var}(X)$

EXAMPLE: CONTINUOUS UNIFORM RV

- ▶ Let X be a continuous RV which chooses a value from $[a, b]$ uniformly at random.

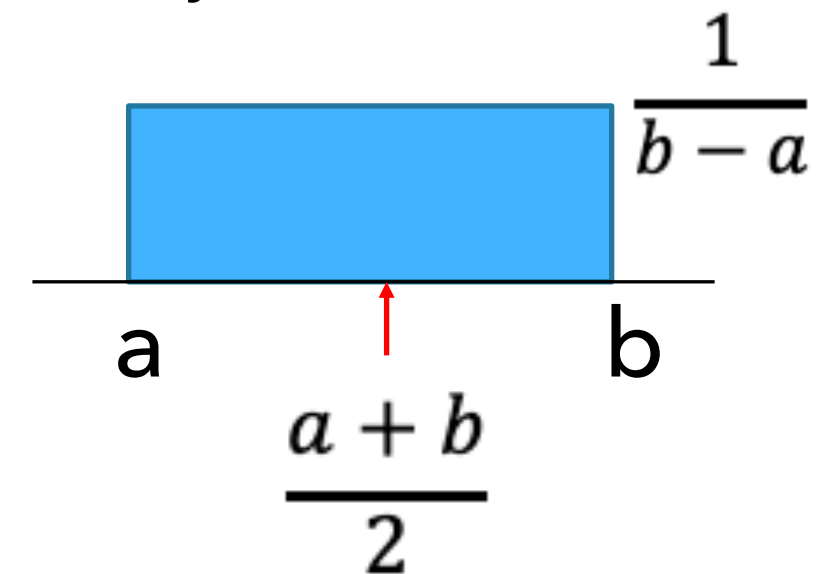
- ▶ The PDF is

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } x \in [a, b] \\ 0 & \text{otherwise} \end{cases}$$

- ▶ We have

$$\mathbb{E}(X) = \frac{a+b}{2}$$

$\text{Var}(X) =$



COMPUTE $\mathbb{E}(X^2)$

PROPERTIES OF VARIANCE

- ▶ By linearity of expectation, we have

$$\mathbb{E}(aX + b) = a \cdot \mathbb{E}(X) + b$$

- ▶ Does this hold for variance? Do we have:

$$\text{Var}(aX + b) = a \cdot \text{Var}(X) + b ?$$

PROPERTIES OF VARIANCE

$$\text{Var}(aX + b) =$$

PROPERTIES OF VARIANCE

- ▶ Again, for expectation by linearity we have for **any** X and Y :

$$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$$

- ▶ Does this hold for variance, such that

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) ?$$

PROPERTIES OF VARIANCE

$$\text{Var}(X + Y) =$$

PROPERTIES OF VARIANCE

- ▶ Let X be a random variable and c be a constant
- ▶ By linearity of expectation, we have

$$\mathbb{E}(cX) = c\mathbb{E}(X) \qquad \mathbb{E}(X + c) = \mathbb{E}(X) + c$$

- ▶ How about $\text{Var}(cX)$?
- ▶ $\text{Var}(X + c)$?

LINEARITY OF VARIANCE?

- There are examples of random variables X and Y such that

$$\text{Var}(X + Y) \neq \text{Var}(X) + \text{Var}(Y)$$

Ω	H	T
$\mathbb{P}$	1/2	1/2
X	1	-1
Y	-1	1
$X + Y$	0	0

$$\text{Var}(X) = \text{Var}(Y) = 1$$

$$\text{Var}(X + Y) = 0$$

PROPERTIES OF VARIANCE

- ▶ Does linearity hold for variance?
- ▶ There are examples of random variables X and Y such that

$$\text{Var}(X + Y) \neq \text{Var}(X) + \text{Var}(Y)$$

- ▶ Equality does hold for **independent** random variables
- ▶ **Theorem:** If $X_1, \dots, X_n$ are **pairwise independent** then

$$\text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n)$$

PROPERTIES OF VARIANCE: SUMMARY

$$\text{Var}(X) = \mathbb{E}((X - \mathbb{E}(X))^2) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

Variance is non-negative: $\text{Var}(X) \geq 0$

For any constant c :

- $\text{Var}(c) = 0$
- $\text{Var}(X + c) = \text{Var}(X)$
- $\text{Var}(cX) = c^2 \text{Var}(X)$

Linearity holds for **pairwise independent** random variables

$$\text{Var}(X_1 + \cdots + X_n) = \text{Var}(X_1) + \cdots + \text{Var}(X_n)$$

DISTRIBUTIONS: MOTIVATION

- ▶ The pdf and cdf of random variables do not involve the sample space
- ▶ There are many random variables arising in different probability spaces with the same pdf/cdf
- ▶ Random variables with the same pdf/cdf behave very similarly
- ▶ Can classify random variables into distributions and study them in a unified way

BERNOULLI DISTRIBUTION

- ▶ Let X be an indicator random variable:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

- ▶ X has the Bernoulli distribution with parameter p
- ▶ Notation: $X \sim \text{Bernoulli}(p)$



Jacob Bernoulli
[1654 - 1705]

BERNOULLI DISTRIBUTION

- ▶ Let X be an indicator random variable:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

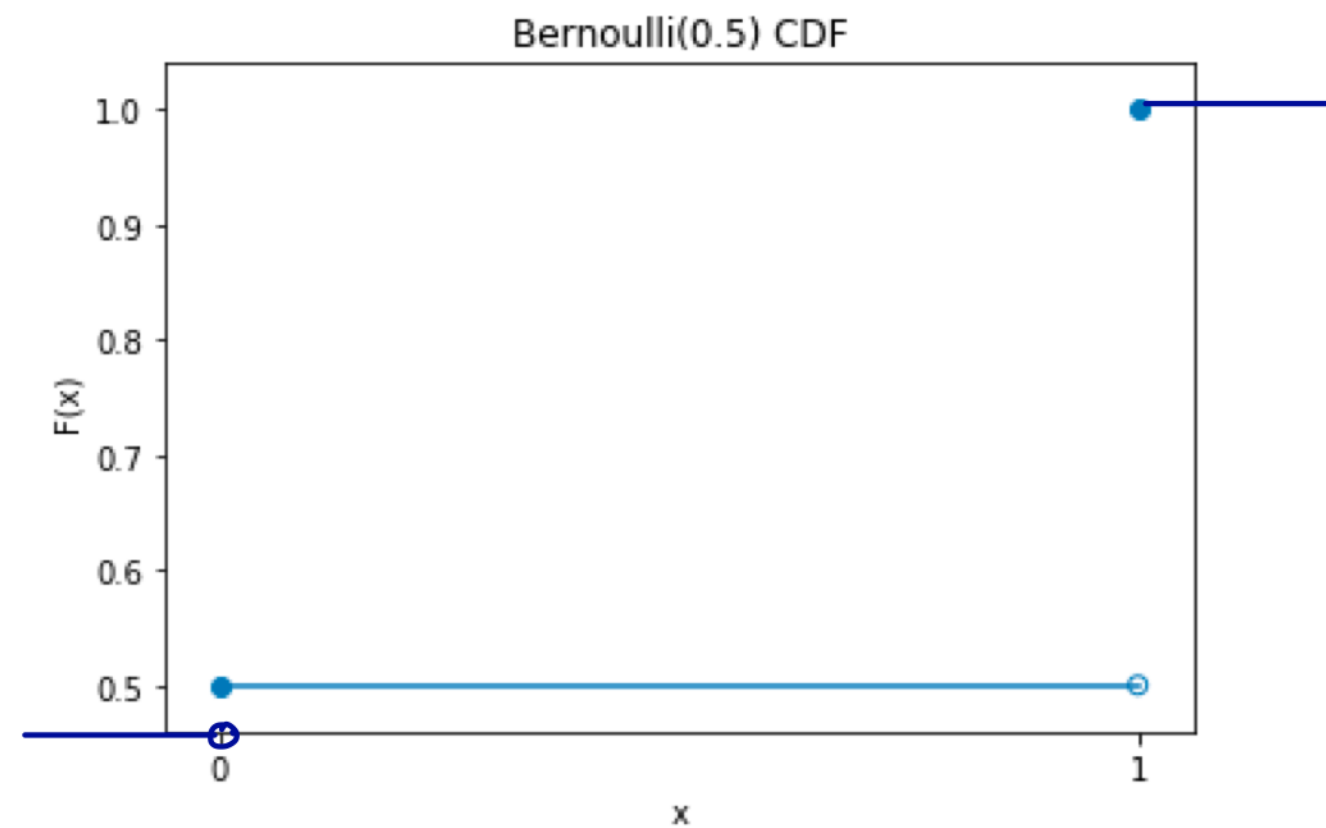
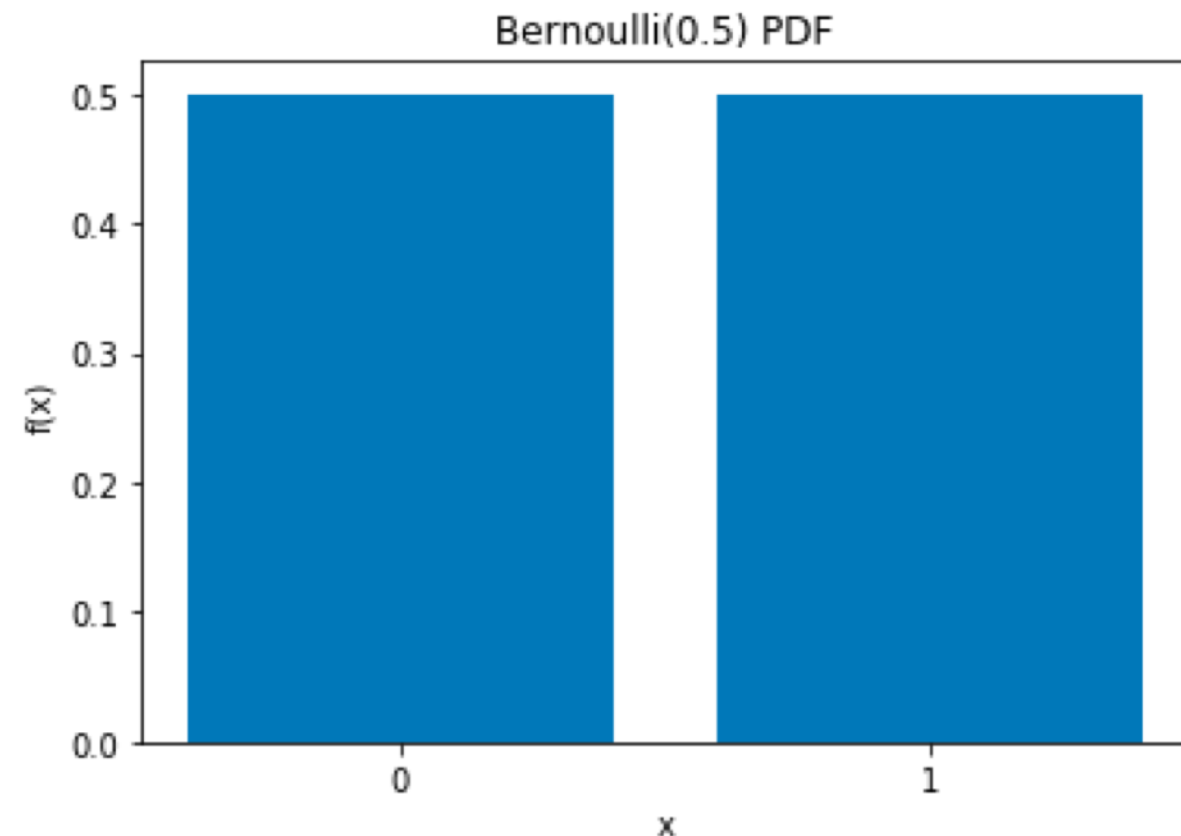
- ▶ The PDF and CDF are:

$$f_p(x) = \begin{cases} 1-p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{otherwise} \end{cases} \quad F_p(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1-p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

BERNOULLI DISTRIBUTION

- Let X be an indicator random variable:

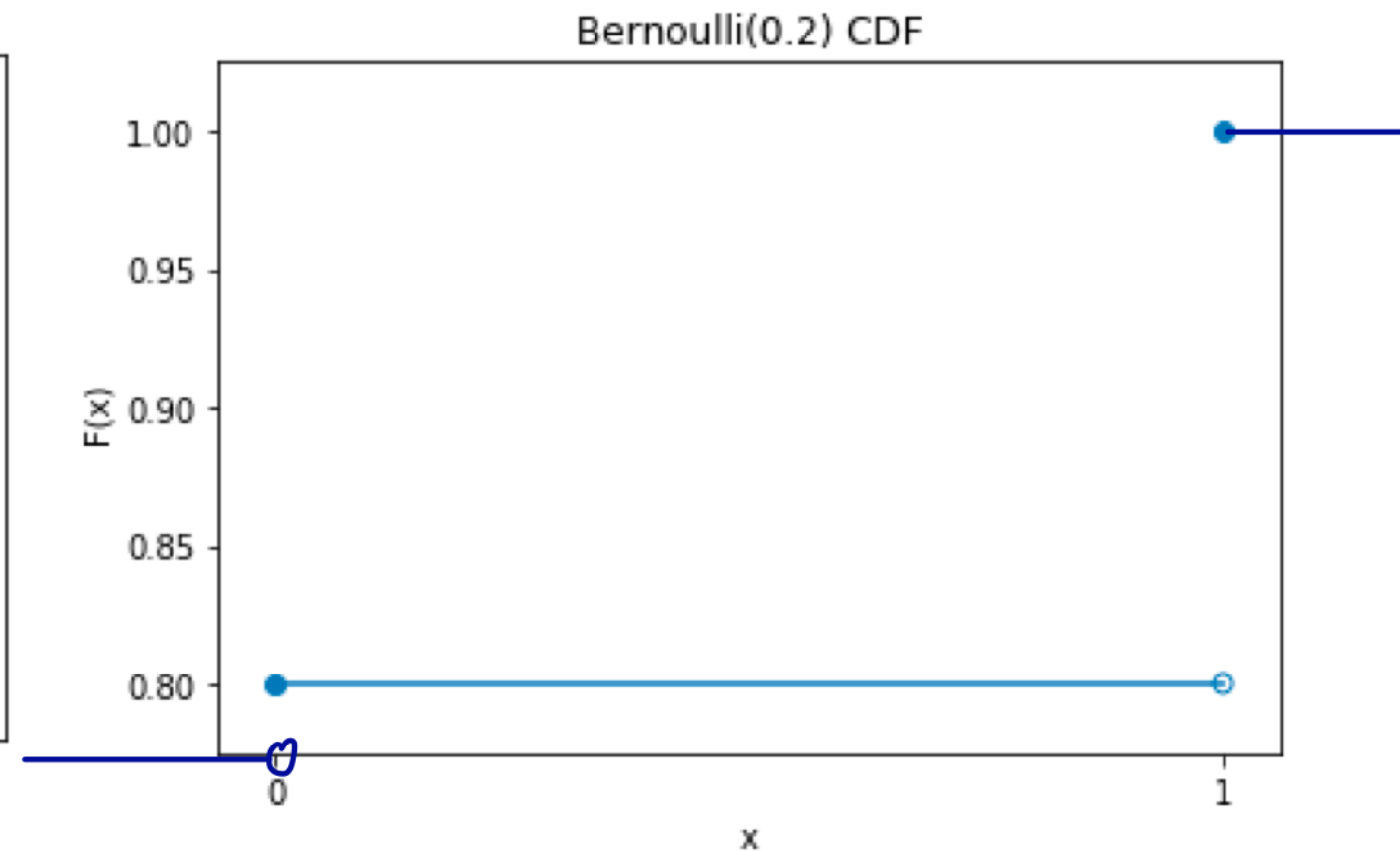
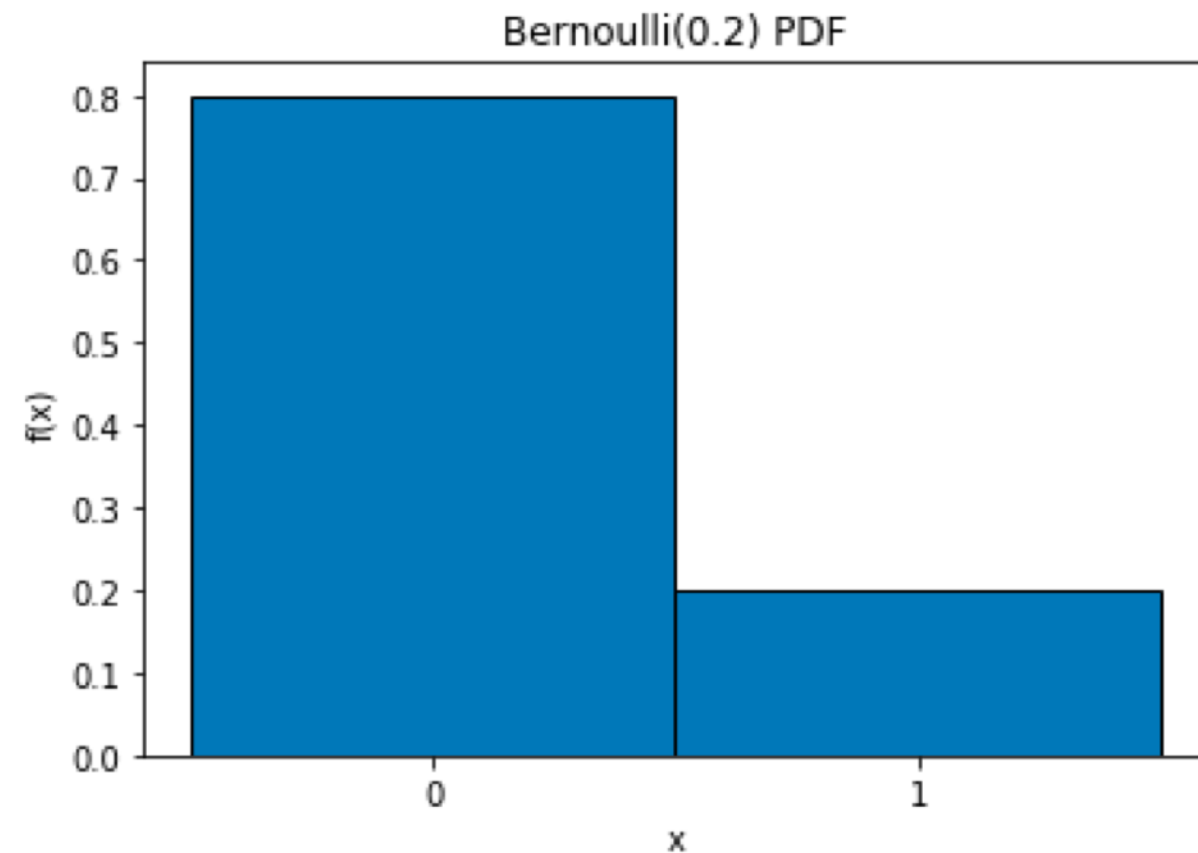
$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$



BERNOULLI DISTRIBUTION

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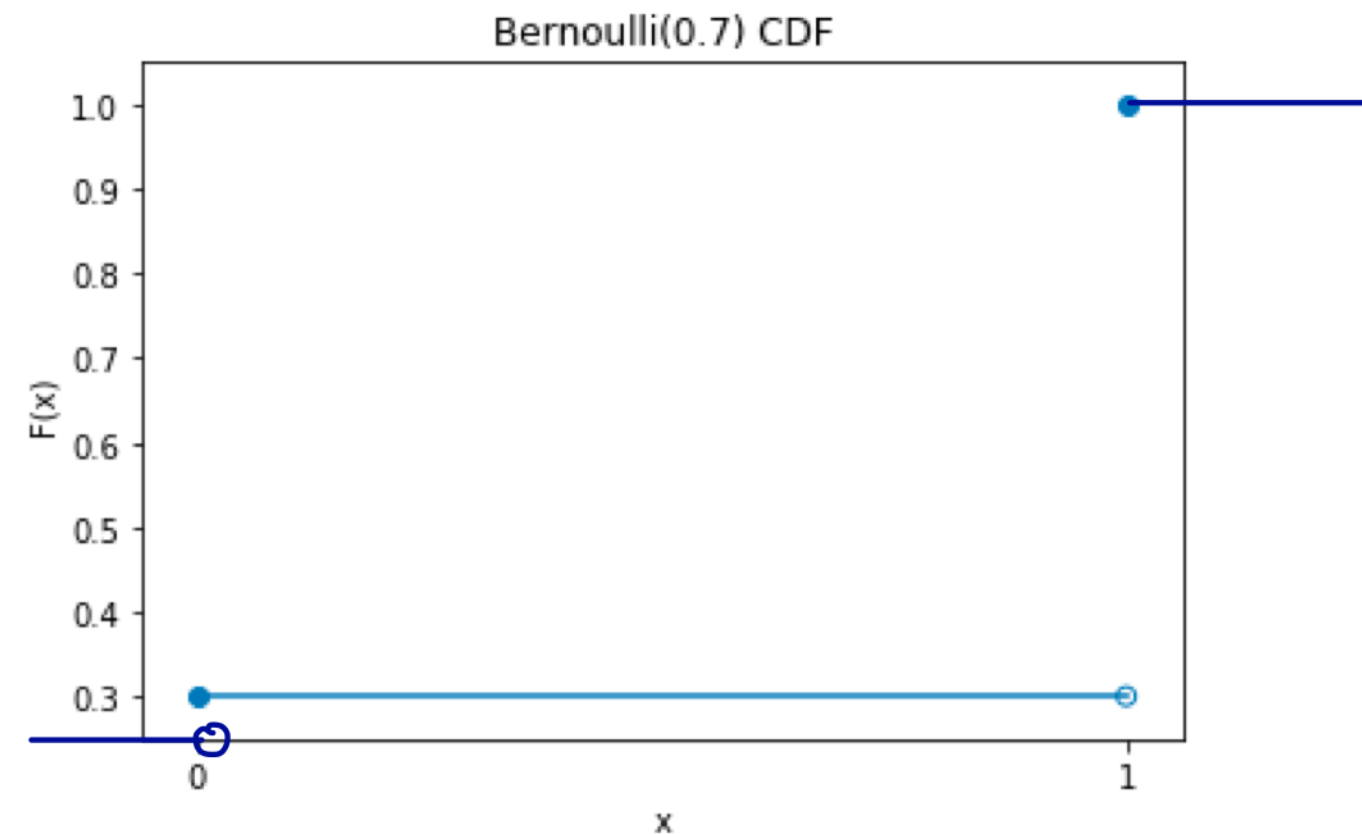
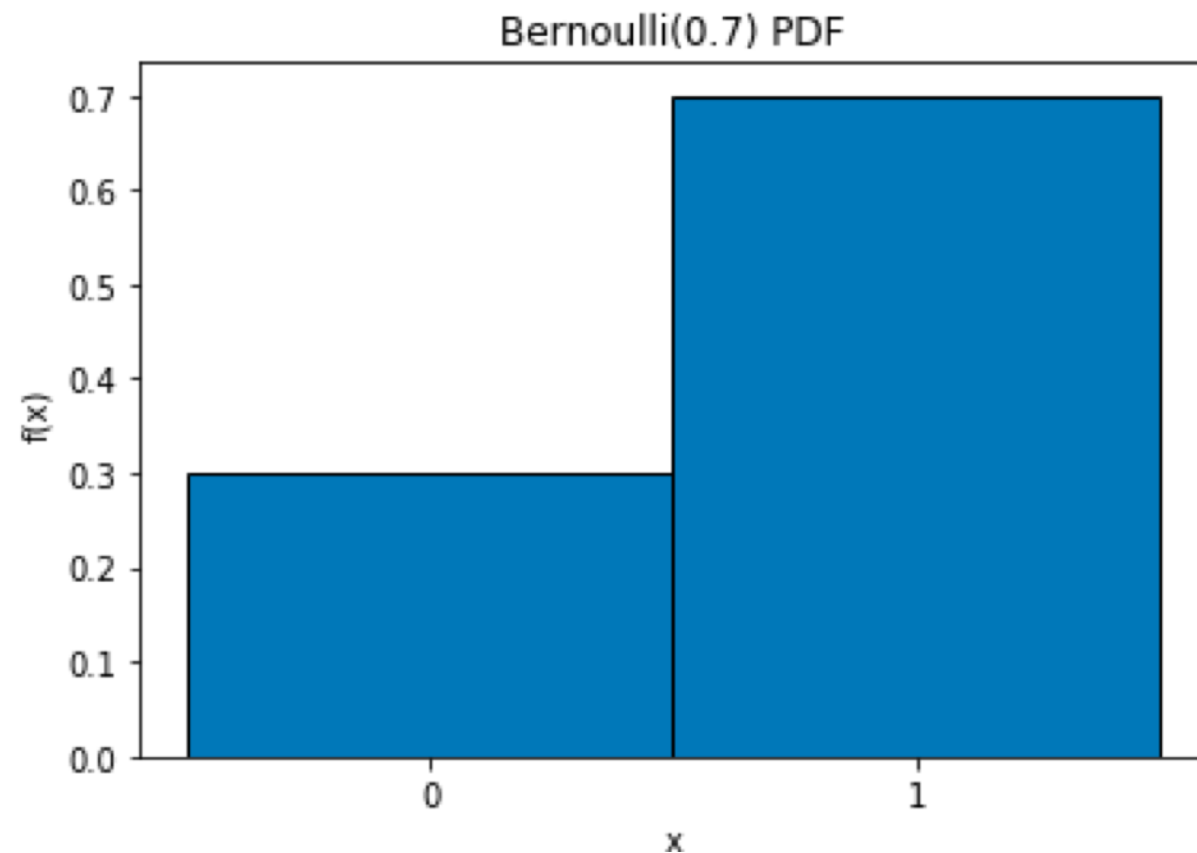
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BERNOULLI DISTRIBUTION

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BERNOULLI DISTRIBUTION

- ▶ Let X be an indicator random variable:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

- ▶ Models any scenario where there is a single trial that succeeds with probability p ($X = 1$ if success)
- ▶ Example: toss a coin with $\Pr(H) = p$ and let $X = 1$ if heads

BERNOULLI DISTRIBUTION

- ▶ Let X be an indicator random variable:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1-p \end{cases}$$

- ▶ Expectation: $\mathbb{E}(X) = p$
- ▶ Variance: $\text{Var}(X) = p(1 - p)$

ASK THE AUDIENCE

- ▶ The variance of a Bernoulli(p) random variable is $p(1 - p)$

Which Bernoulli r.v. has **maximum** variance?

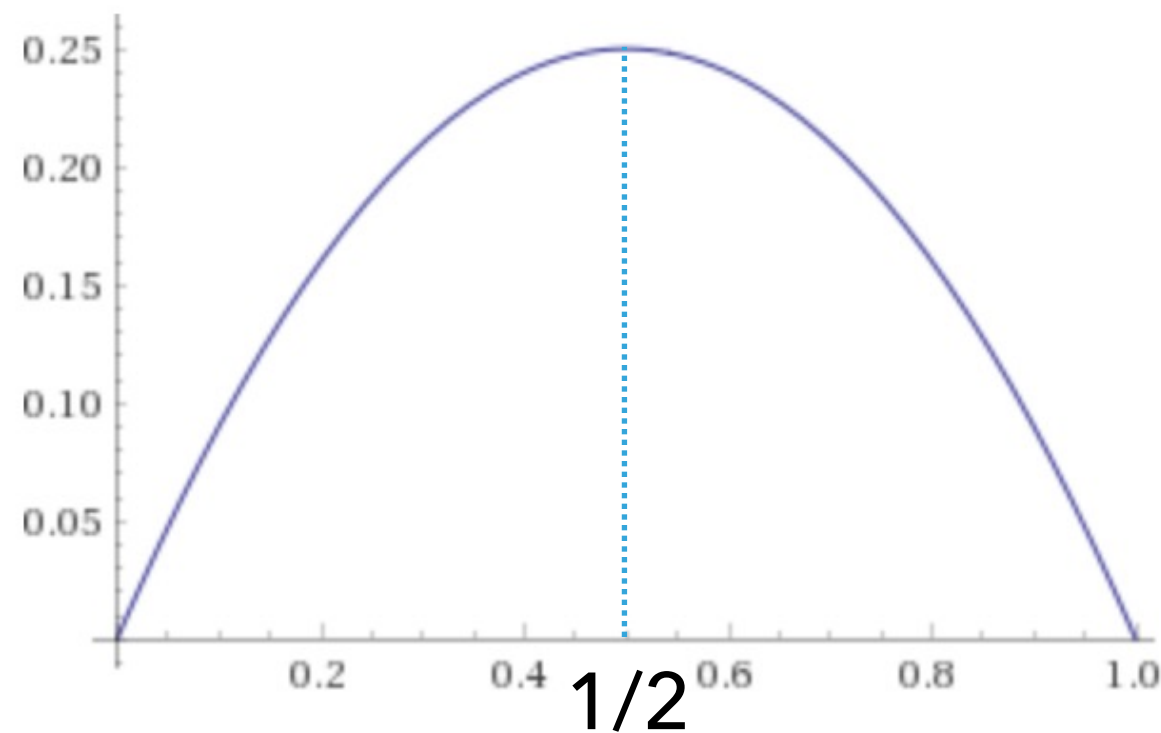
(A) $p = 1$ (B) $p = 1/2$ (C) $p = 2/3$ (D) none of the above

ASK THE AUDIENCE

- ▶ The variance of a Bernoulli(p) random variable is $p(1 - p)$

Which Bernoulli r.v. has **maximum** variance?

(A) $p = 1$ (B) $p = 1/2$ (C) $p = 2/3$ (D) none of the above



DISCRETE UNIFORM DISTRIBUTION

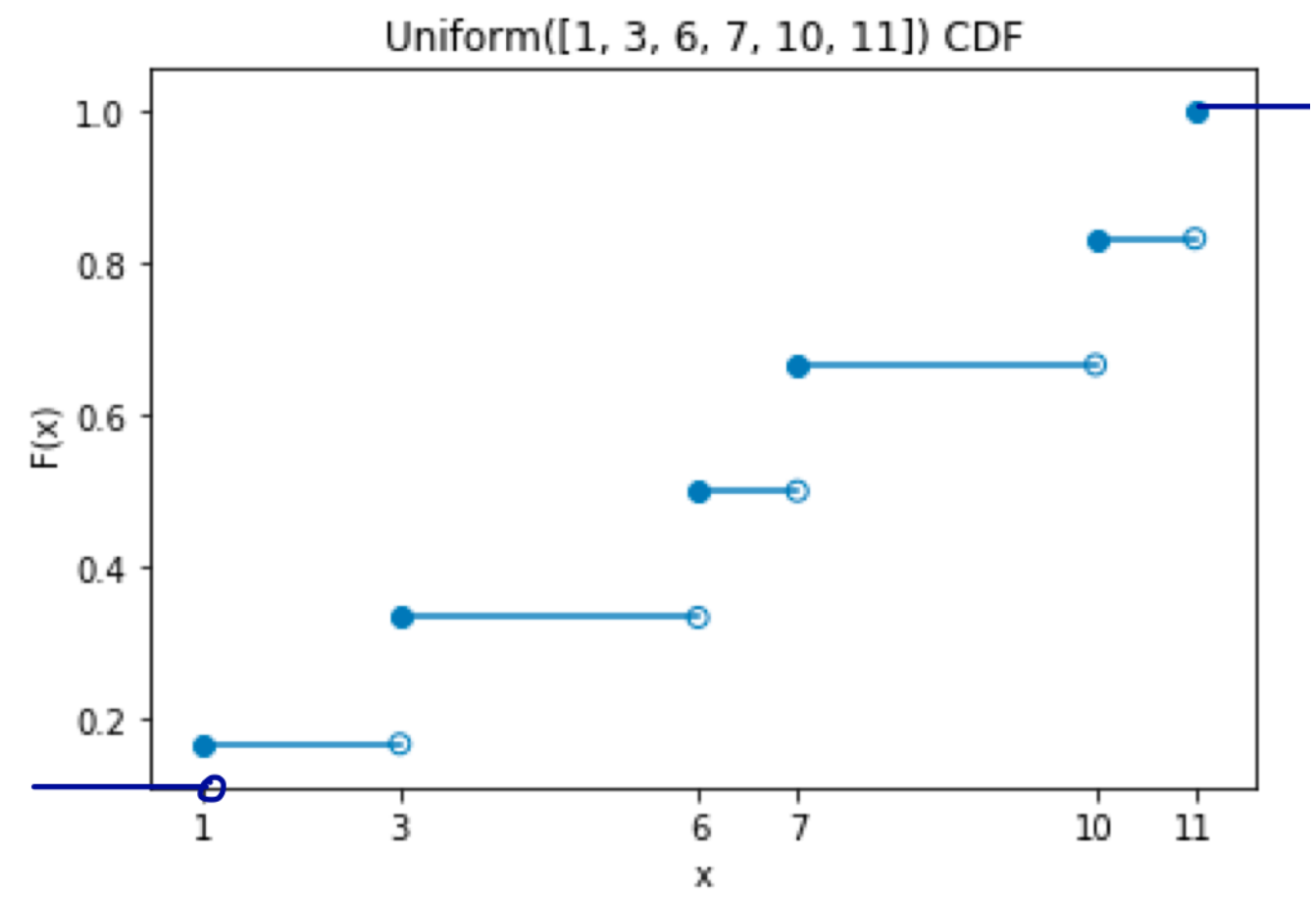
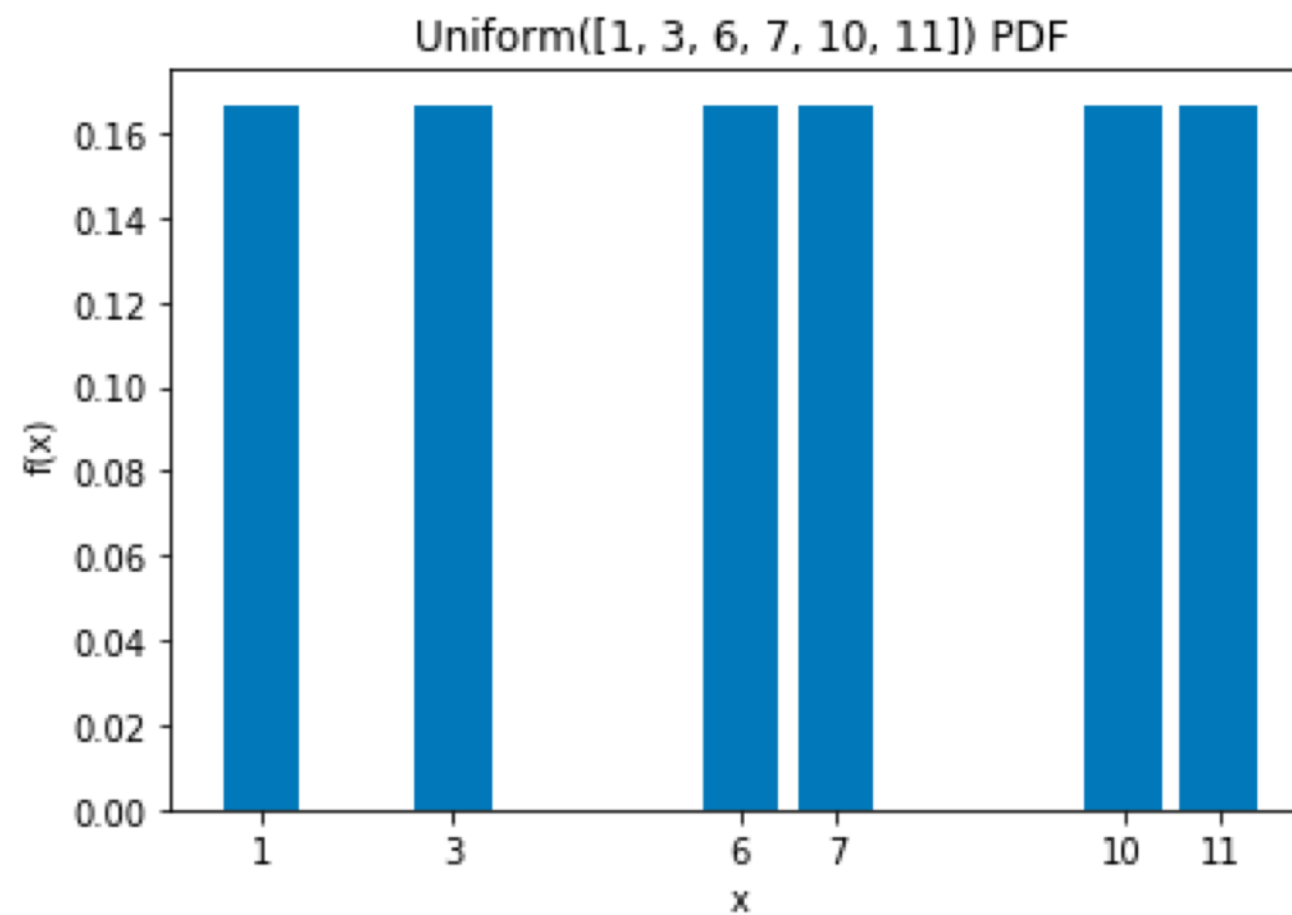
- ▶ Let S be a finite set of values: $s_1 < s_2 < \dots < s_n$
- ▶ A random variable X that is uniform over S has PMF/CDF:

$$f_X(x) = \begin{cases} 1/n & \text{if } x \text{ in } S \\ 0 & \text{if } x \notin S \end{cases}$$

$$F_X(x) = \begin{cases} 0 & \text{if } x < s_1 \\ k/n & \text{if } s_k \leq x < s_{k+1} \text{ for } 1 \leq k < n. \\ 1 & \text{if } x \geq s_n \end{cases}$$

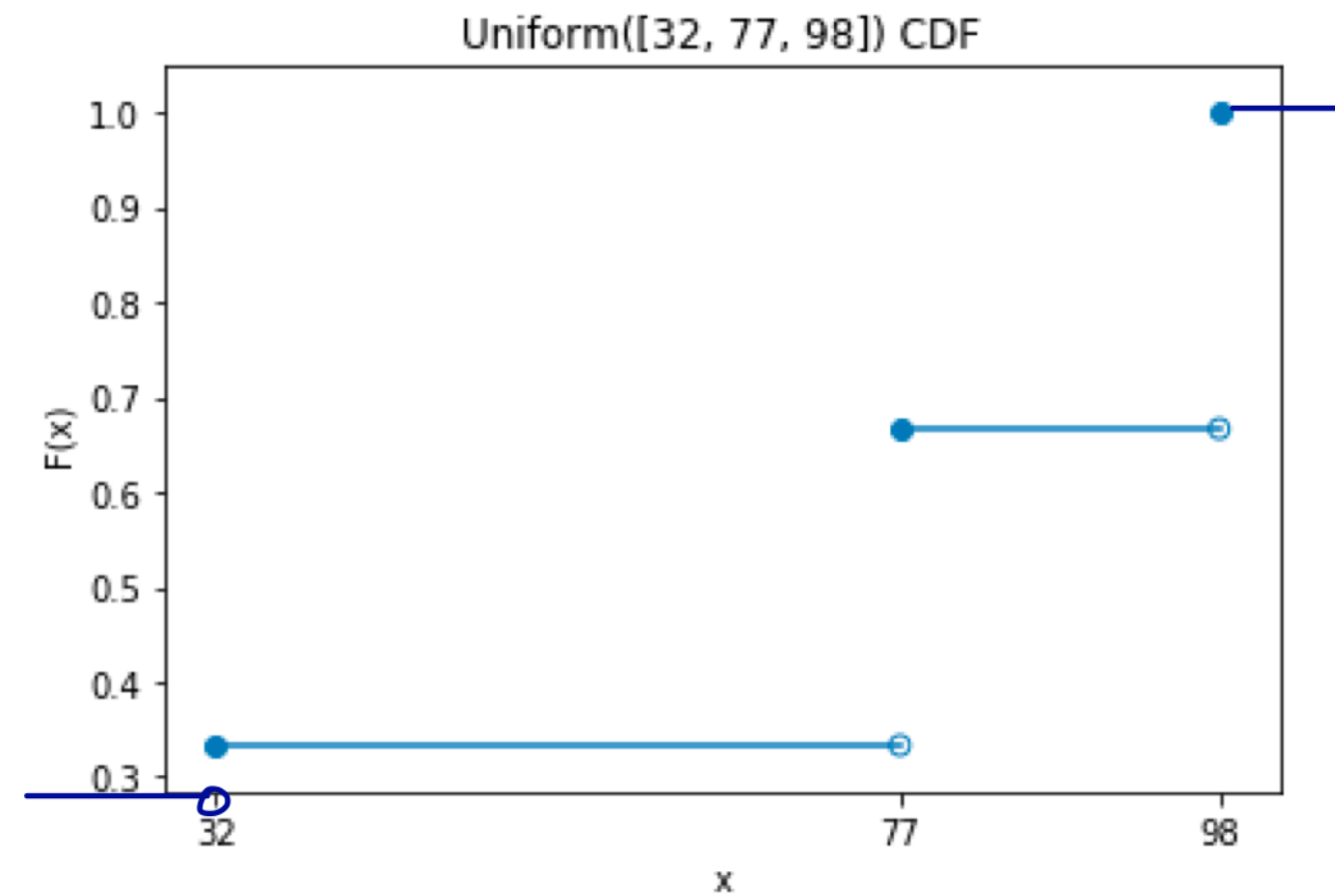
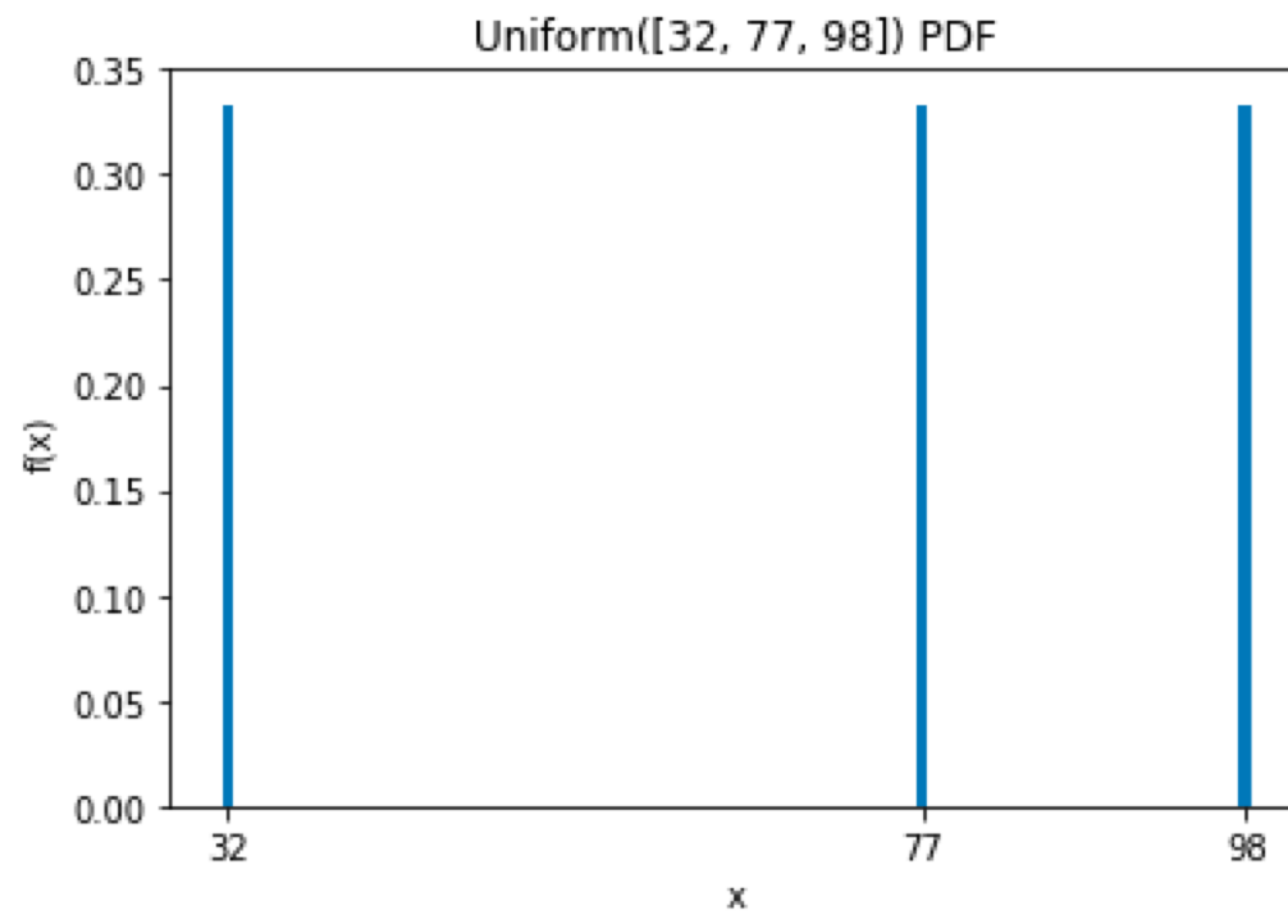
DISCRETE UNIFORM DISTRIBUTION

- ▶ $X \sim \text{Uniform}(S)$ examples:



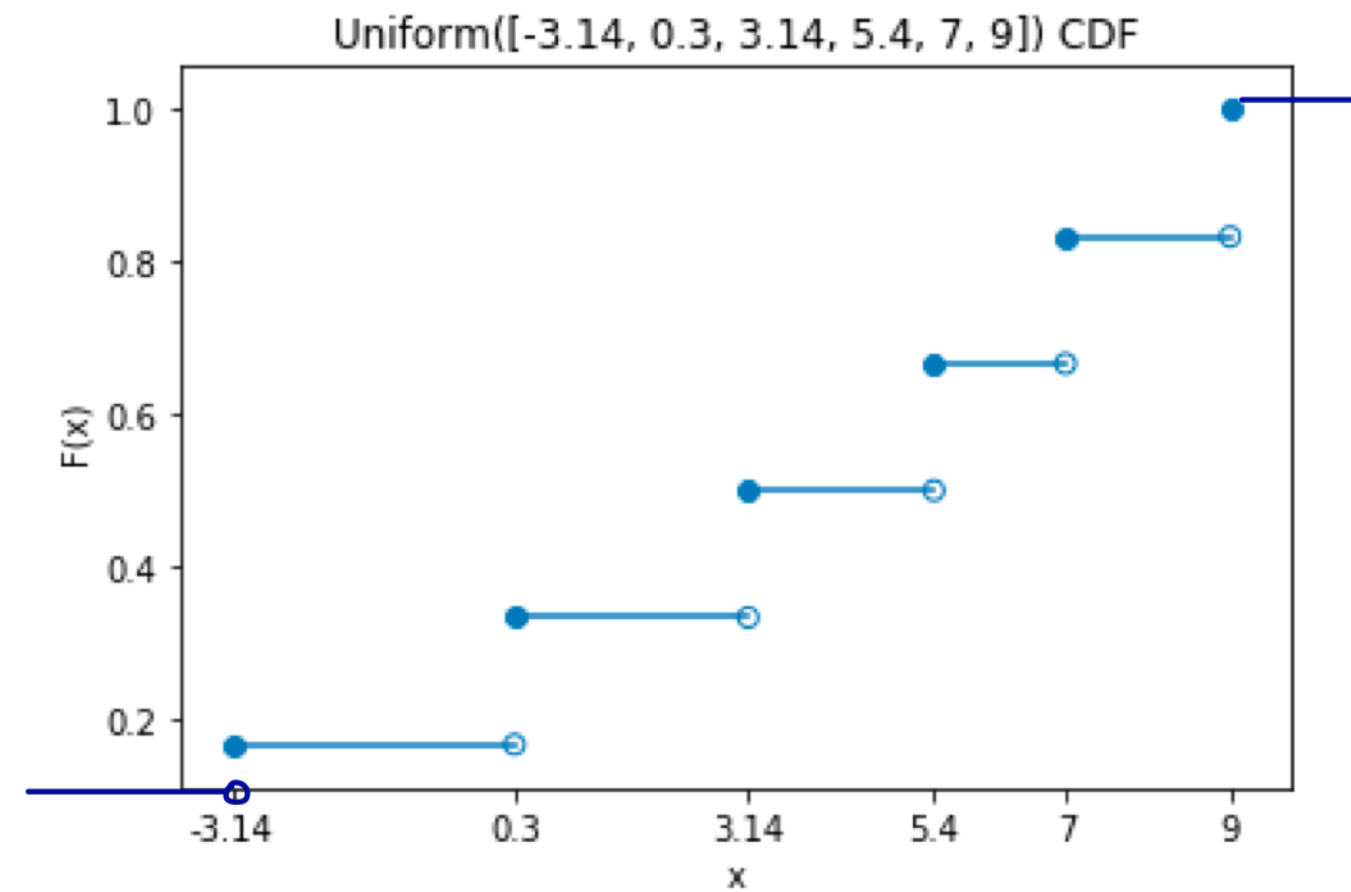
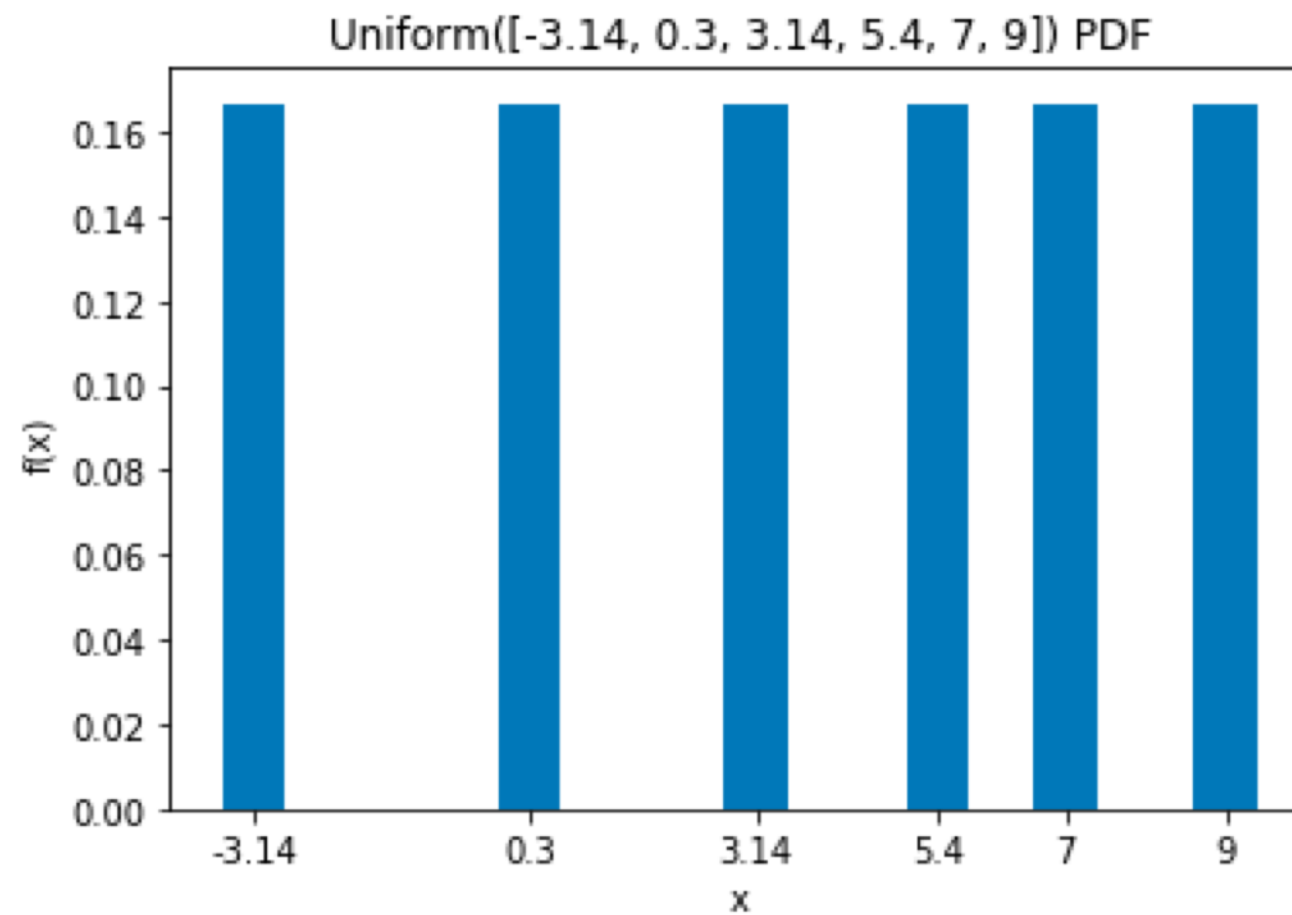
DISCRETE UNIFORM DISTRIBUTION

- ▶ $X \sim \text{Uniform}(S)$ examples:



DISCRETE UNIFORM DISTRIBUTION

- ▶ $X \sim \text{Uniform}(S)$ examples:



DISCRETE UNIFORM DISTRIBUTION

- ▶ Let $X \sim \text{Uniform}(\{s_1, s_2, \dots, s_n\})$
- ▶ Expectation and variance:

$$\mathbb{E}(X) = \sum_{i=1}^n s_i \cdot \frac{1}{n} = \frac{s_1 + s_2 + \dots + s_n}{n}$$

$$\mathbb{E}(X^2) = \sum_{i=1}^n s_i^2 \cdot \frac{1}{n} = \frac{s_1^2 + s_2^2 + \dots + s_n^2}{n}$$

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{s_1^2 + s_2^2 + \dots + s_n^2}{n} - \frac{(s_1 + s_2 + \dots + s_n)^2}{n^2}$$

DISCRETE UNIFORM DISTRIBUTION

- ▶ Example: $X \sim \text{Uniform}(\{1, 2, \dots, n\})$
- ▶ We have the following closed form solutions:

$$\mathbb{E}(X) = \frac{\sum_{i=1}^n i}{n} = \frac{n+1}{2}$$

$$\mathbb{E}(X^2) = \frac{\sum_{i=1}^n i^2}{n} = \frac{(n+1)(2n+1)}{6}$$

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \frac{(n-1)(n+1)}{12}$$

UNIFORM DISCRETE DISTRIBUTION

- ▶ $X \sim \text{Uniform}(a, b)$

$$f(x) = \begin{cases} \frac{1}{N} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

for $N = b - a + 1$

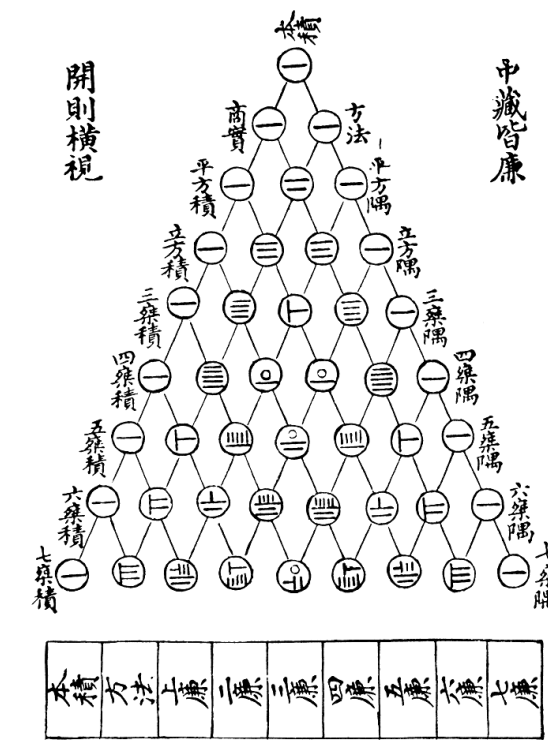
- ▶ $\mathbb{E}(X) = \frac{a + b}{2}$

- ▶ $\text{Var}(X) = \frac{N^2 - 1}{12}$

BINOMIAL DISTRIBUTION

- ▶ n independent trials
- ▶ Each trial is a success with probability p
- ▶ X = number of successes

$$Pr(X = k) =$$

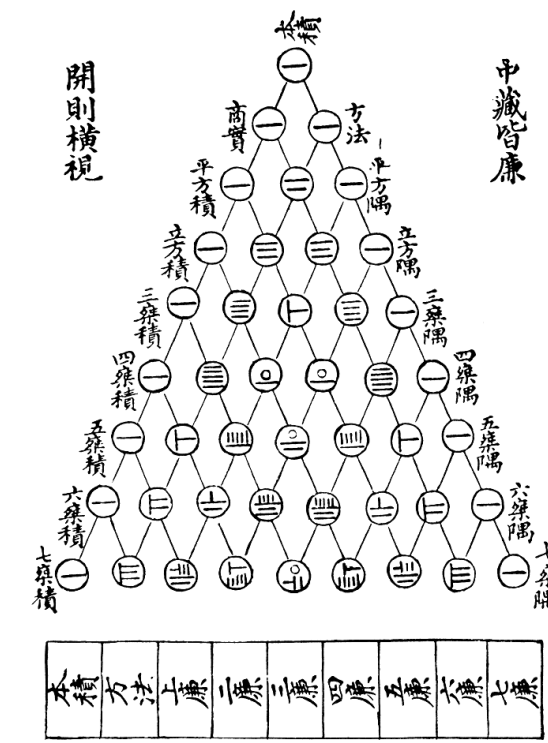


Yang Hui's triangle
(Pascal's triangle)

BINOMIAL DISTRIBUTION

- ▶ n independent trials
- ▶ Each trial is a success with probability p
- ▶ X = number of successes

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$



Yang Hui's triangle
(Pascal's triangle)

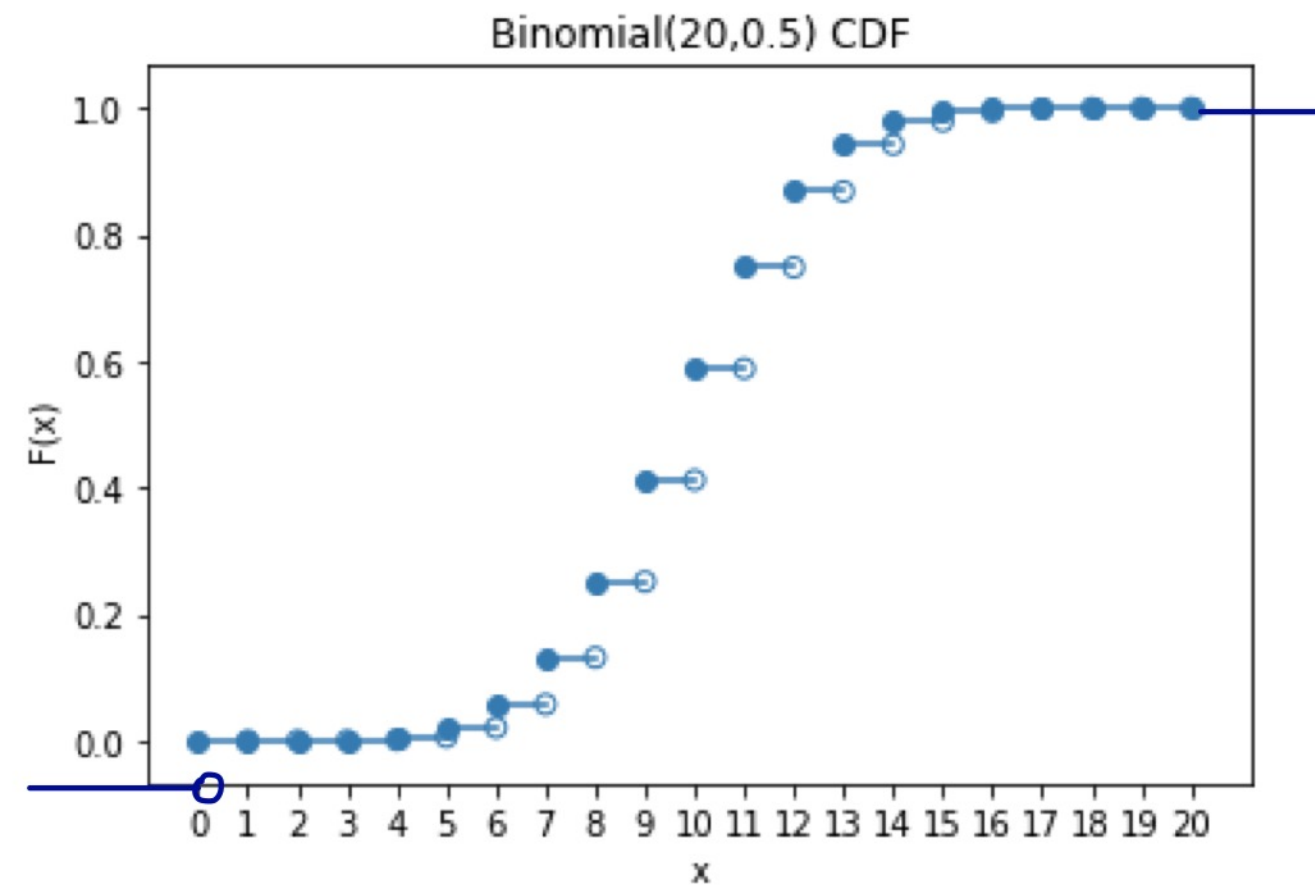
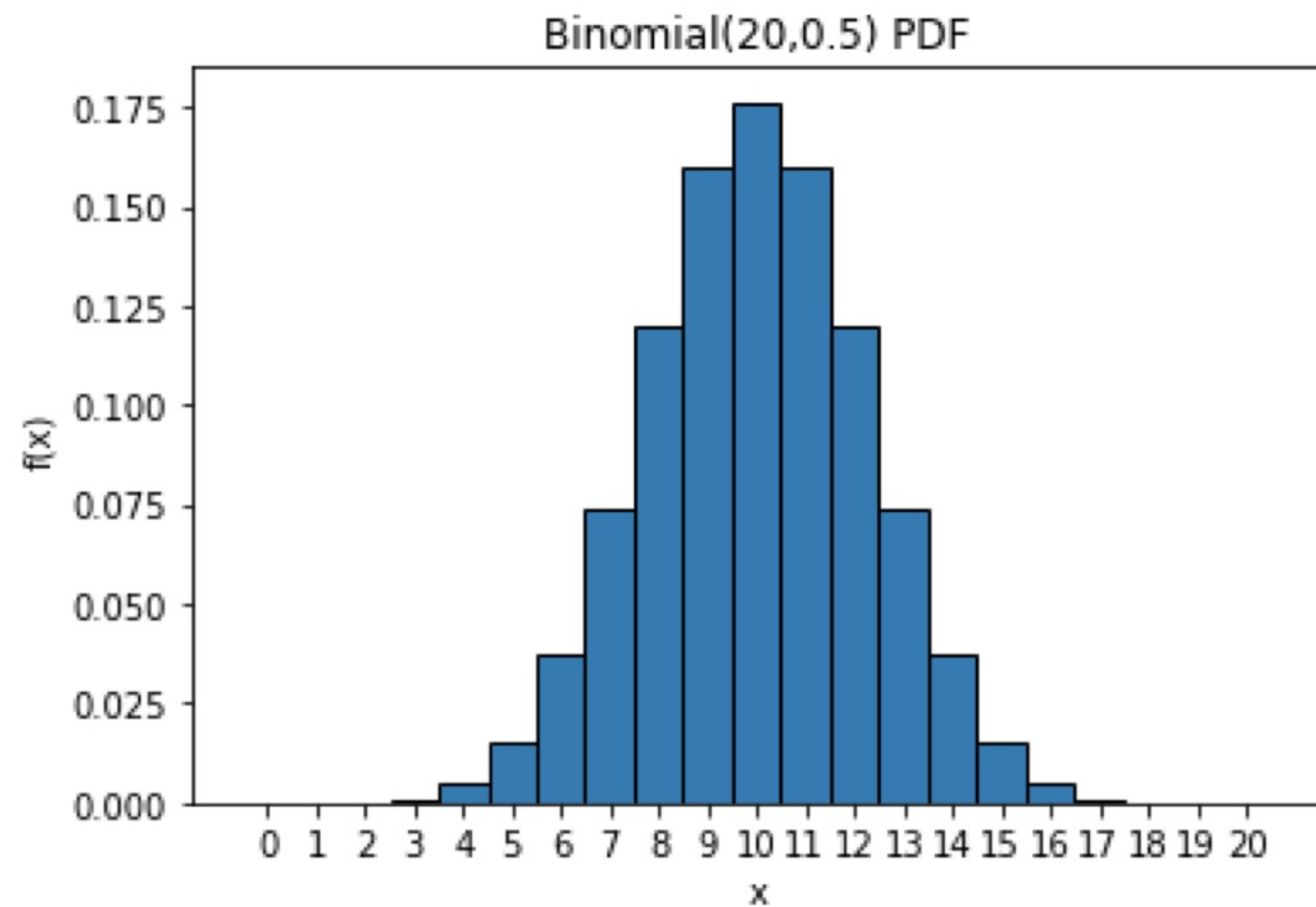
BINOMIAL DISTRIBUTION

- ▶ The random experiment behind the binomial distribution is as follows. Suppose that I have a coin with $\Pr(H) = p$.
- ▶ I toss the coin **N times** and define X to be the **total number of heads** that I observe.
- ▶ Then X is binomial with parameter N and p , and we write $X \sim \text{Binomial}(N, p)$ with $R_X = \{0, 1, 2, \dots, N\}$.

$$\Pr(X = k) = \binom{N}{k} \cdot p^k \cdot (1 - p)^{N-k}$$

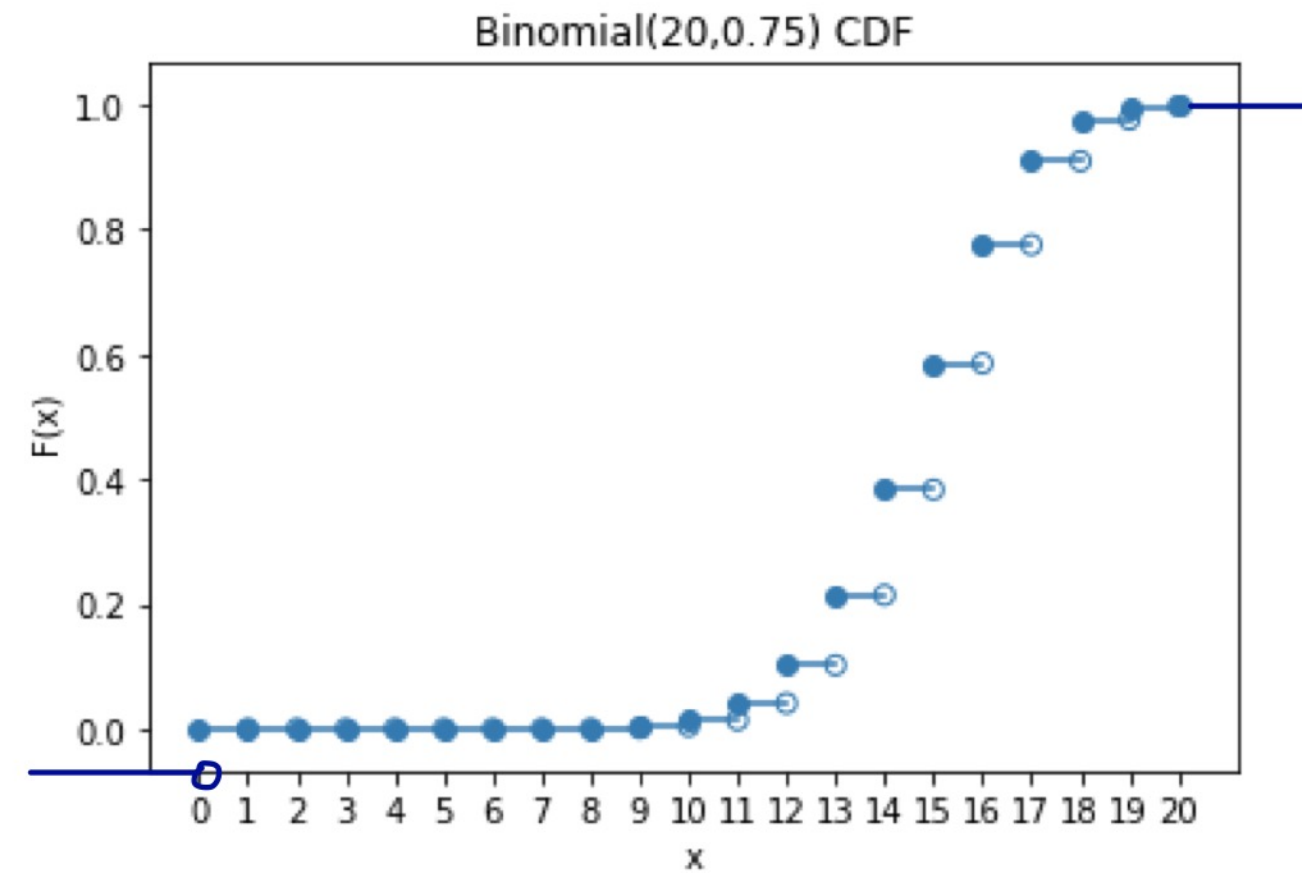
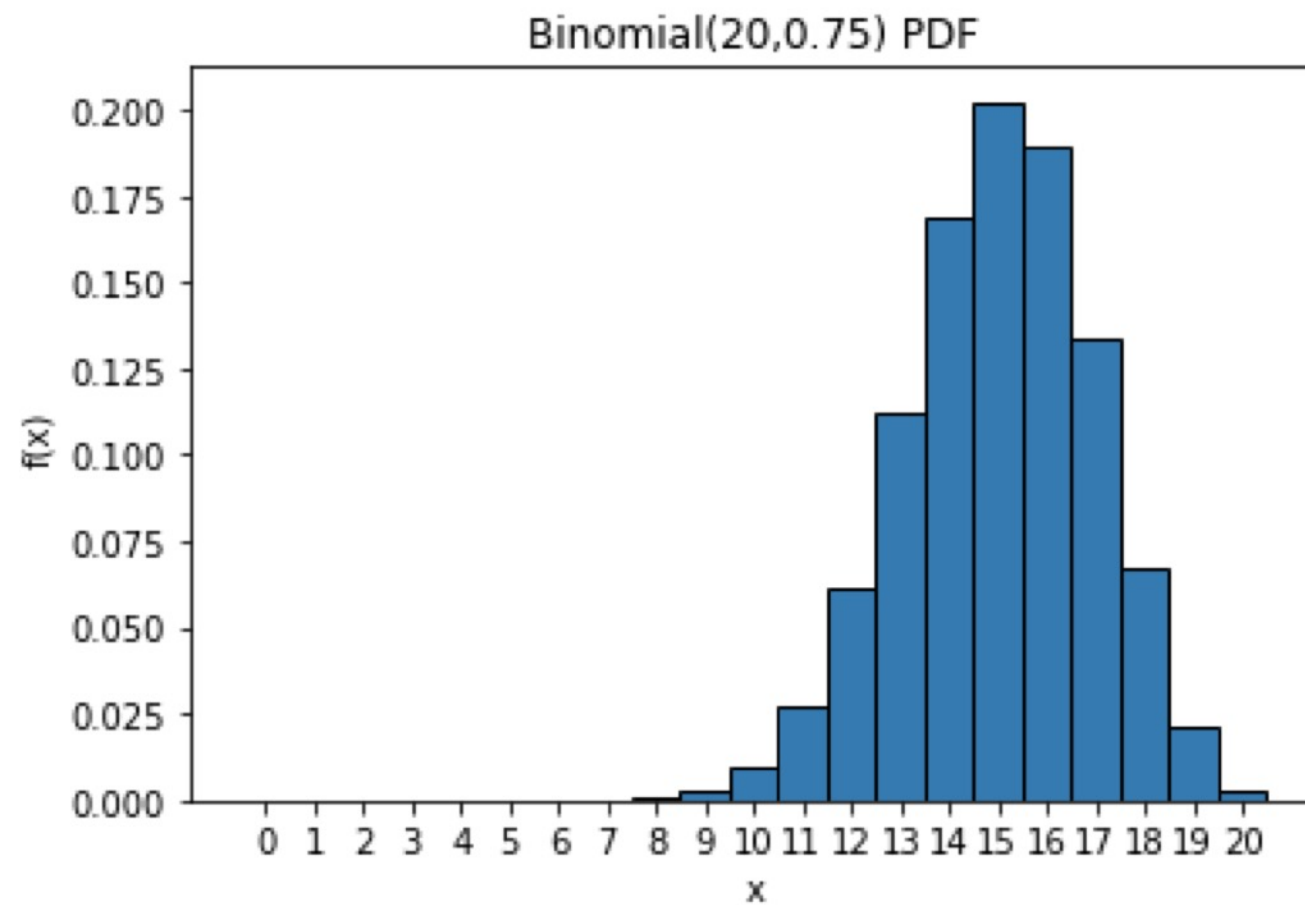
BINOMIAL DISTRIBUTION

- ▶ $X \sim \text{Binomial}(n, p)$ example



BINOMIAL DISTRIBUTION

- ▶ $X \sim \text{Binomial}(n, p)$ example



BINOMIAL DISTRIBUTION

- ▶ Let $X \sim \text{Binomial}(n, p)$
- ▶ Expectation:

ASK THE AUDIENCE

Let $X \sim \text{Binomial}(n, p)$.

What is $\text{Var}(X)$?

A. $n \cdot p$

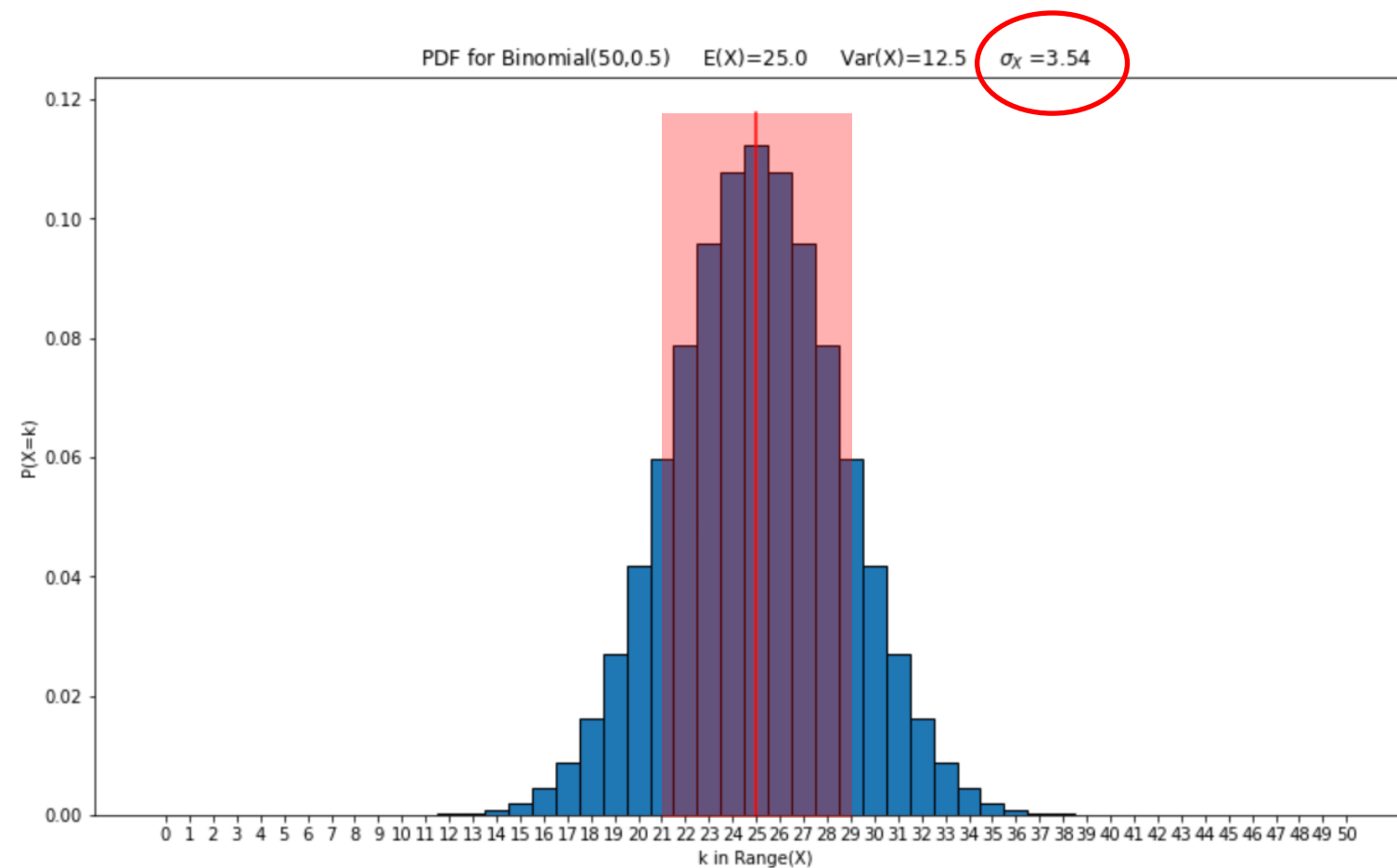
B. $n \cdot p(1 - p)$

C. $p(1 - p)$

D. None of the above

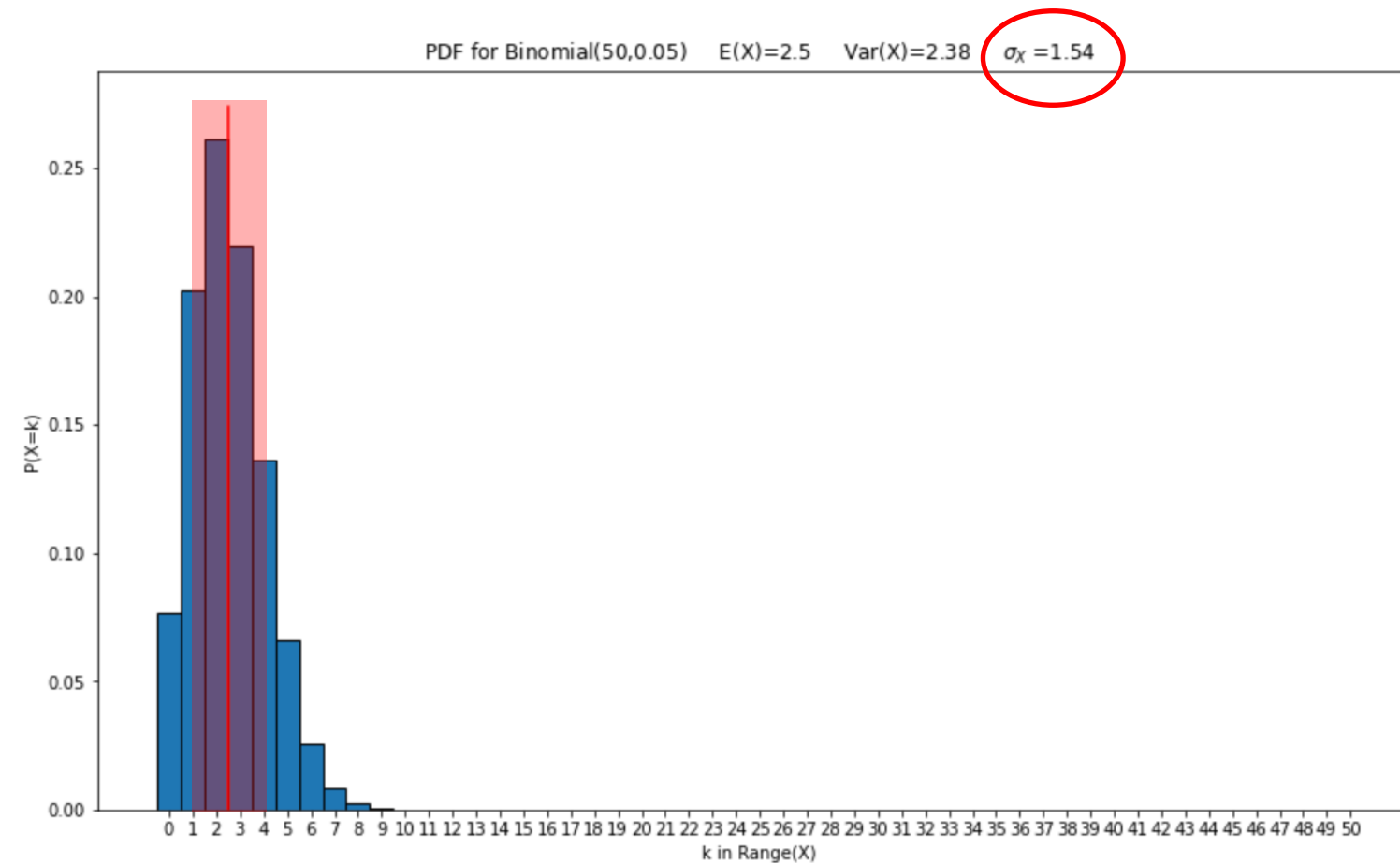
BINOMIAL DISTRIBUTION

- ▶ Let $X \sim \text{Binomial}(n, p)$
- ▶ $\text{Var}(X) = n \cdot p \cdot (1 - p)$



BINOMIAL DISTRIBUTION

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BINOMIAL DISTRIBUTION

- ▶ Let $X \sim \text{Binomial}(n, p)$
- ▶ Variance:

BINOMIAL DISTRIBUTION

- ▶ Let $X \sim \text{Binomial}(n, p)$
- ▶ $\text{Var}(X) = n \cdot p \cdot (1 - p)$

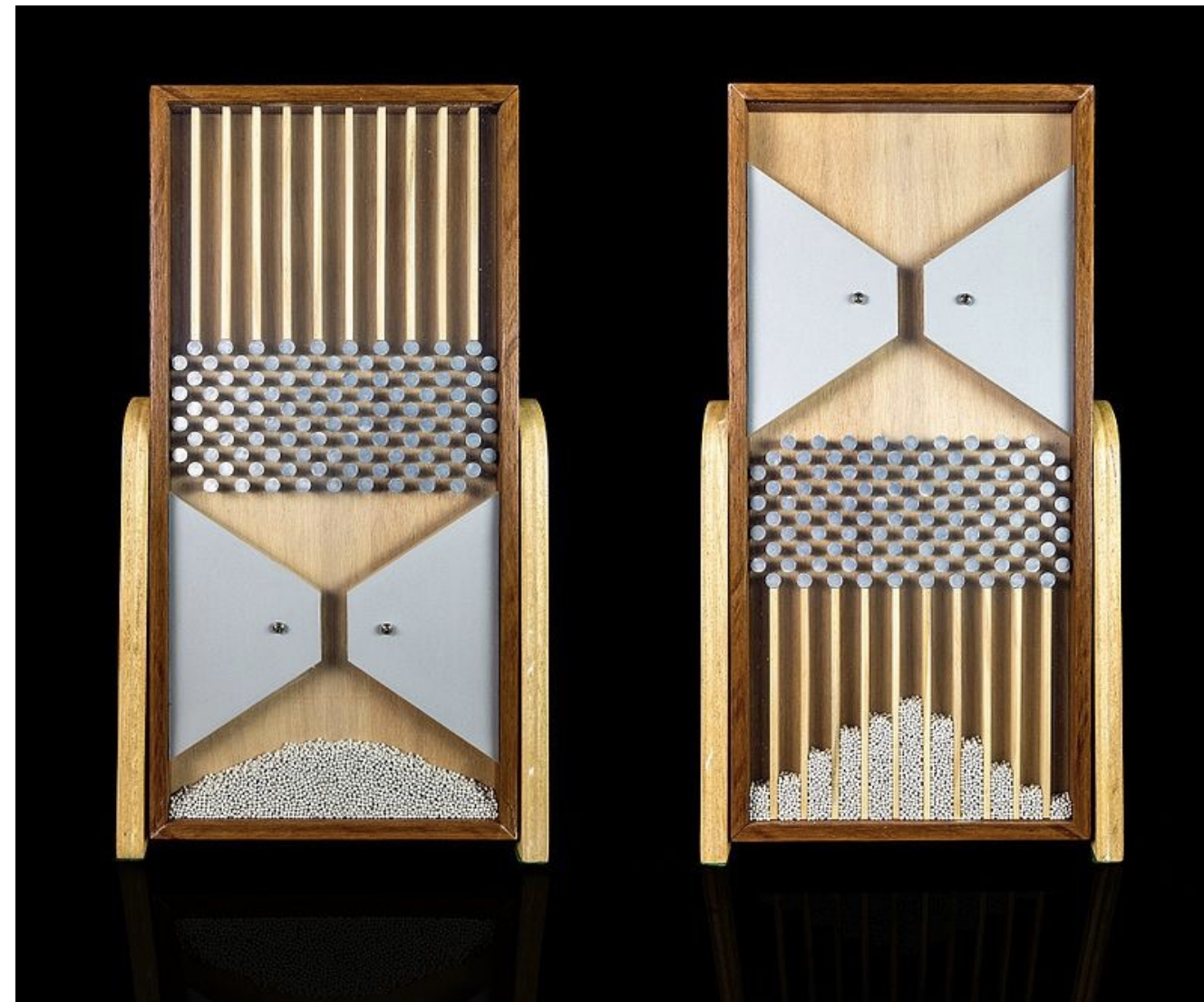
The variance of a sum of **independent random variables** is the sum of the variances.

PEOPLE V. COLLINS REVISITED

- ▶ For a real-life application of binomial probabilities, please see the People v. Collins example from Lecture 12
- ▶ In that example, we had n couples ($n = 5,000,000$)
- ▶ Each couple had a probability p of matching the witnesses' description, independently of others ($p = 1/12,000,000$)
- ▶ The random variable of interest was:
 $X = \text{number of couples matching the description}$
- ▶ We now know that $X \sim \text{Binomial}(n, p)$

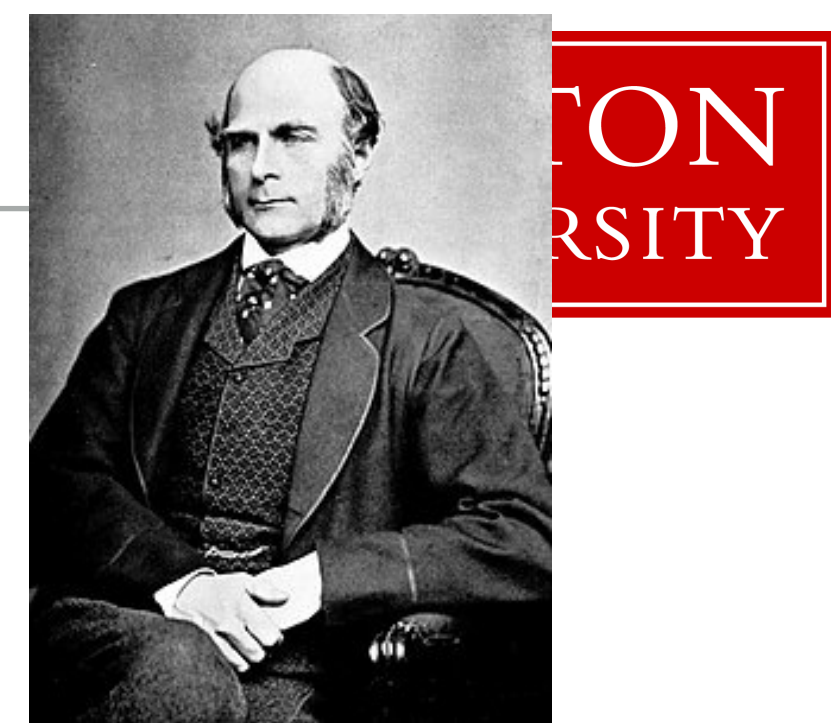
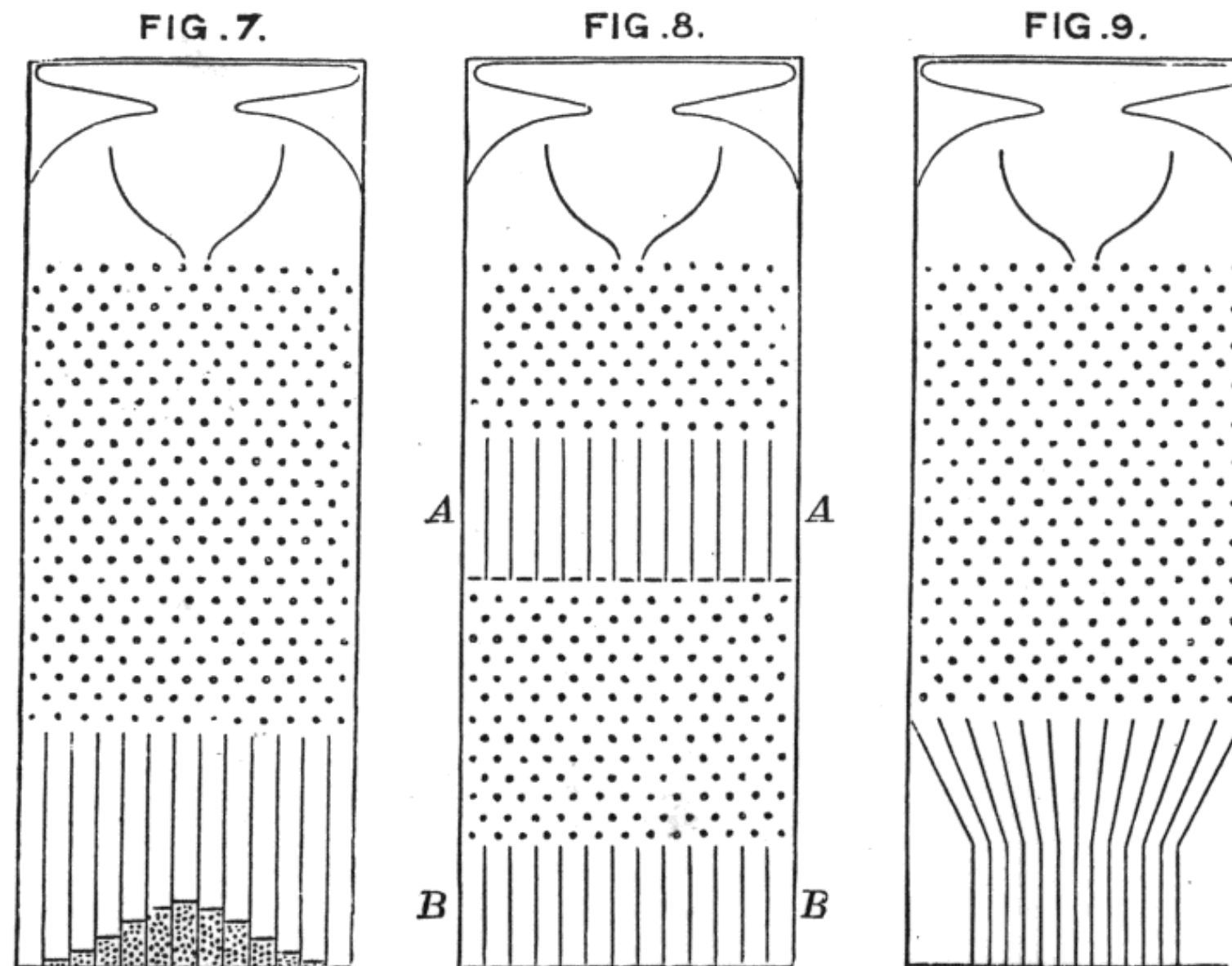
GALTON BOARD

- ▶ The Galton board (aka the bean machine or quincunx)



GALTON BOARD

- ▶ The Galton board (as drawn by Galton)



Sir Francis Galton
[1822 - 1911]

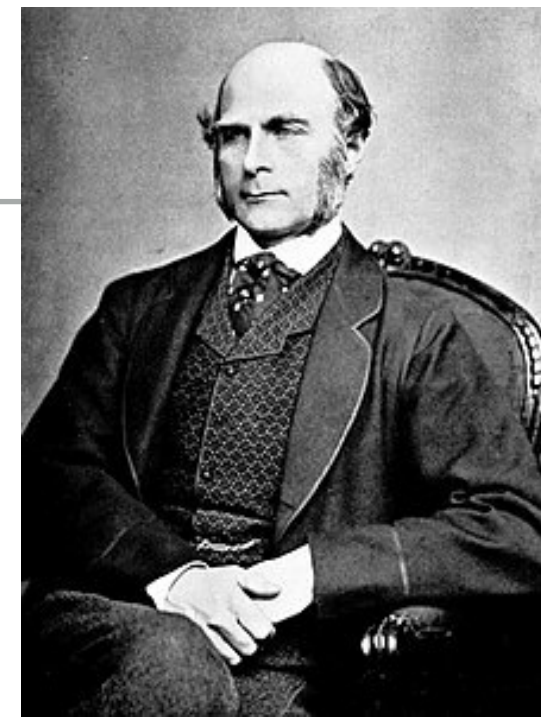
GALTON BOARD

- ▶ The Galton board in action:

https://en.wikipedia.org/wiki/Bean_machine

<https://www.mathsisfun.com/data/quincunx.html>

- ▶ Many Youtube videos on it too



Sir Francis Galton
[1822 - 1911]



GALTON BOARD

- ▶ A natural mathematical model for the Galton board process is the following
- ▶ Let n = number of rows of pegs
- ▶ When a ball hits a peg, it moves either left or right
- ▶ By symmetry, it is natural to model $\Pr(\text{left}) = \Pr(\text{right}) = 1/2$
- ▶ Under this model, we have

$$\Pr(\text{ball ends up in } k\text{-th bin}) = \binom{n}{k} \frac{1}{2^n}$$

- ▶ The real-life simulations show this model is pretty accurate

GALTON BOARD

"Sir Francis Galton was fascinated with the order of the bell curve that emerges from the apparent chaos of beads bouncing off of pegs in the Galton Board. He eloquently described this relationship in his book *Natural Inheritance*:

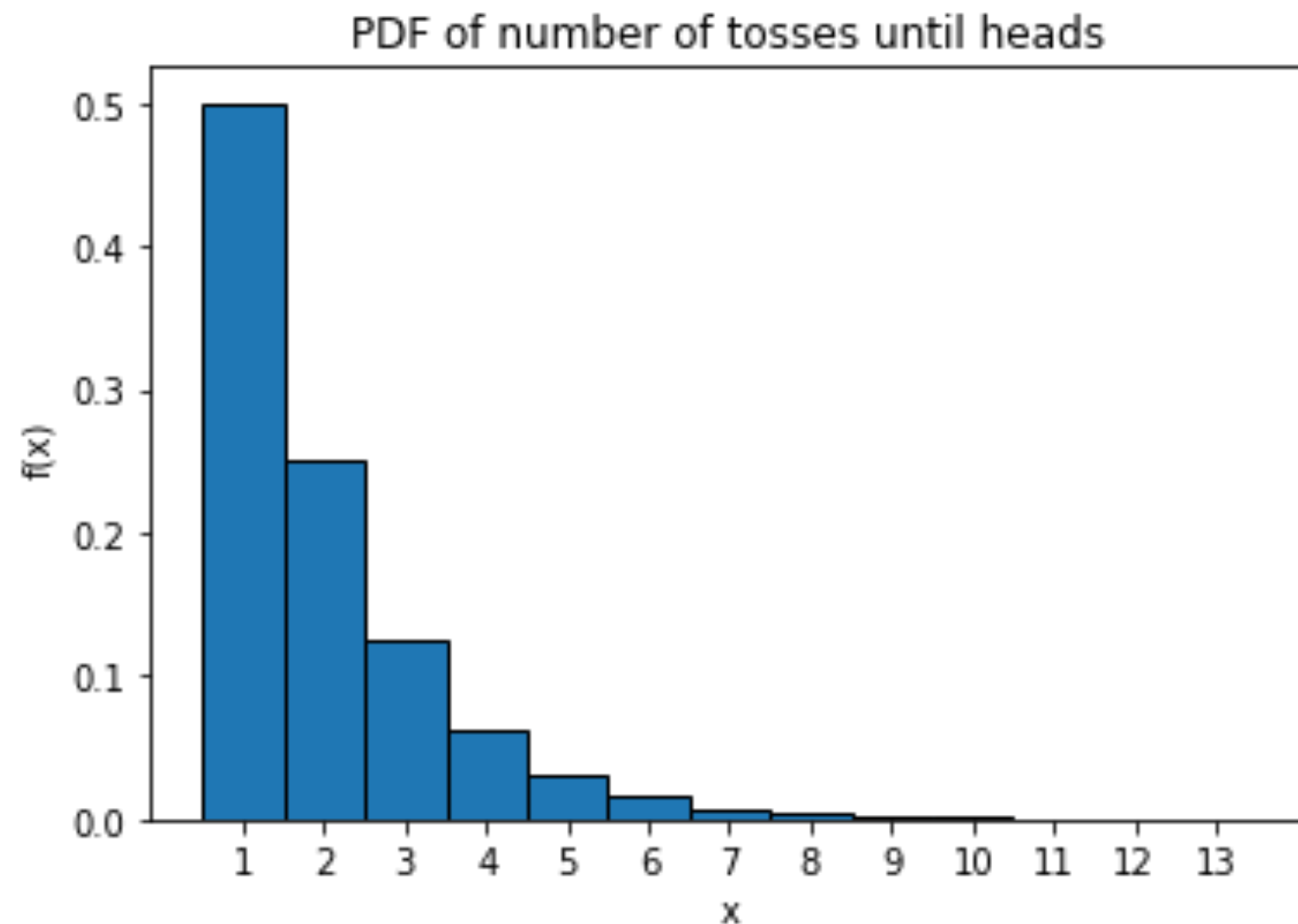
Order in Apparent Chaos: I know of scarcely anything so apt to impress the imagination as the wonderful form of cosmic order expressed by the Law of Frequency of Error. The law would have been personified by the Greeks and deified, if they had known of it. It reigns with serenity and in complete self-effacement amidst the wildest confusion. The huger the mob, and the greater the apparent anarchy, the more perfect is its sway. It is the supreme law of Unreason. Whenever a large sample of chaotic elements are taken in hand and marshalled in the order of their magnitude, an unsuspected and most beautiful form of regularity proves to have been latent all along."

Quoted from https://en.wikipedia.org/wiki/Bean_machine

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LECTURE 17

Last time

- Variance properties
- Discrete distributions I

Today

- Discrete distributions II